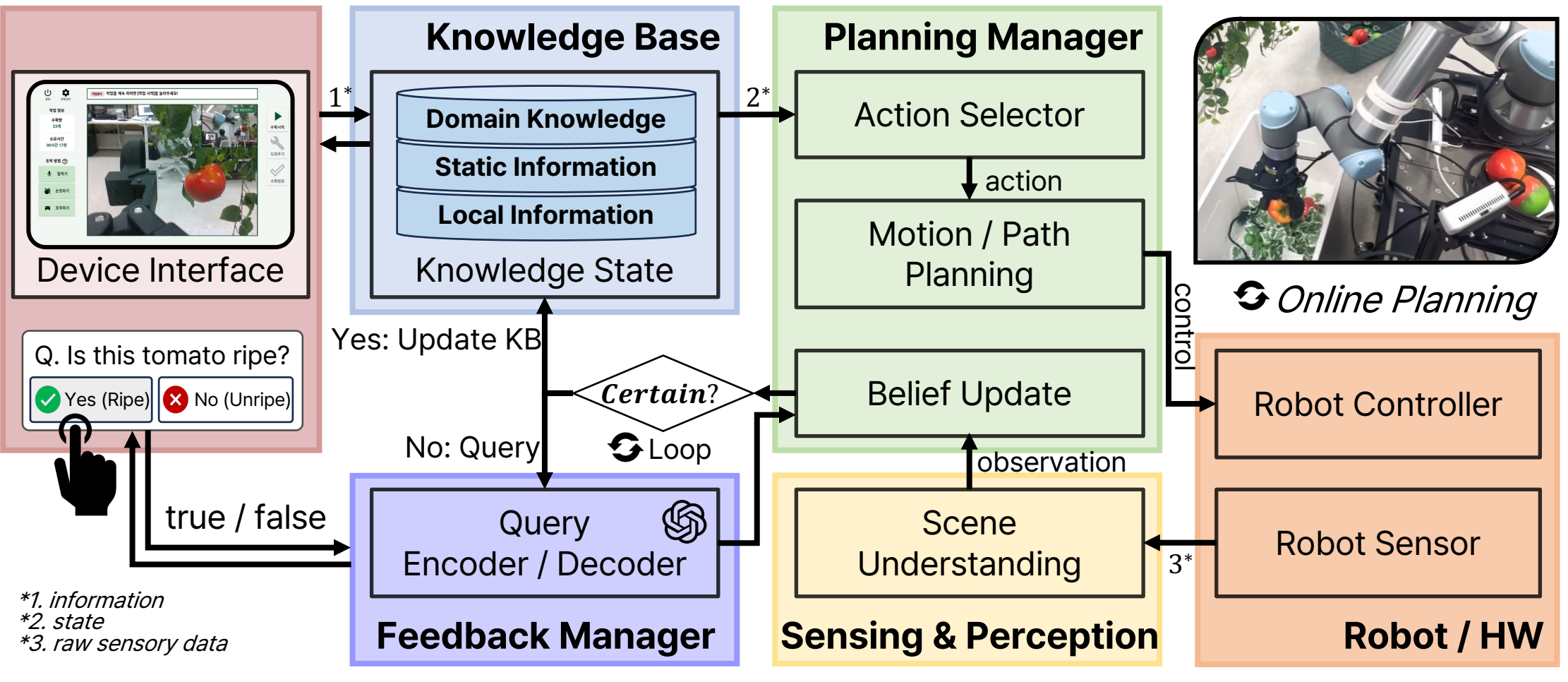

Knowledge Base
Domain Knowledge
Static Information
Local Information
Knowledge State
Planning Manager
Action Selector
action
Motion / Path Planning
Belief Update
Online Planning
Device Interface
Q. Is this tomato ripe?
Yes (Ripe)
No (Unripe)
1*
2*
control
Yes: Update KB
Certain?
Loop
No: Query
observation
true / false
Query Encoder / Decoder
Feedback Manager
Scene Understanding
Sensing & Perception
3*
Robot Controller
Robot Sensor
Robot / HW
*1. information
*2. state
*3. raw sensory data